

Build a Manufacturing-Data App with Gemini

Use Gemini to understand the Majestic Interiors CRM data, then build and refine a working dashboard app with Canvas - no coding (slides 11-17).

Activity is designed by Su Nguyen; Detailed guide is drafted by Claude Cowork.

Before you start

This activity runs in Gemini (gemini.google.com), not SPARK - sign in with your Google account. The 7 Majestic Interiors CRM CSV files are in the Week 7 > data folder (crm_customers, crm_interactions, delivery_installation, design, inventory_log, manufacturing, sales). Review the AI-for-Manufacturing intro video on the Course Canvas first for example prompts, and pick one analytics area (e.g. sales) to focus on.

Tool	Gemini (gemini.google.com) with Canvas
Data	7 Majestic Interiors CRM CSV files
Step A	Attach the CSVs and ask Gemini to describe each dataset's fields
Step B	Ask Gemini (in Canvas) to build a sales dashboard app from the data
Step C	Refine the app by asking - change a chart, add a natural-language filter

Companion recordings (Week7_Gemini_DataOverview.gif, Week7_Gemini_BuildDashboard.gif, Week7_Gemini_FineTune.gif, Week7_Gemini_NLFeature.gif) show every click in sequence.

Step 1 - Understand the data: field descriptions (Slides 11-12)

Before analysing anything, understand what is in the data. Majestic Interiors is a custom-interiors manufacturer; the dataset spans customers, interactions, design, manufacturing, inventory, delivery and sales.

1. In Gemini, click the + button and choose Upload files, then attach all 7 CRM CSV files from the Week 7 > data folder.
2. Ask: "Give a table describing the fields for each dataset attached."
3. Gemini returns a clear field-by-field table for every file - field name, data type, and a plain-English description.

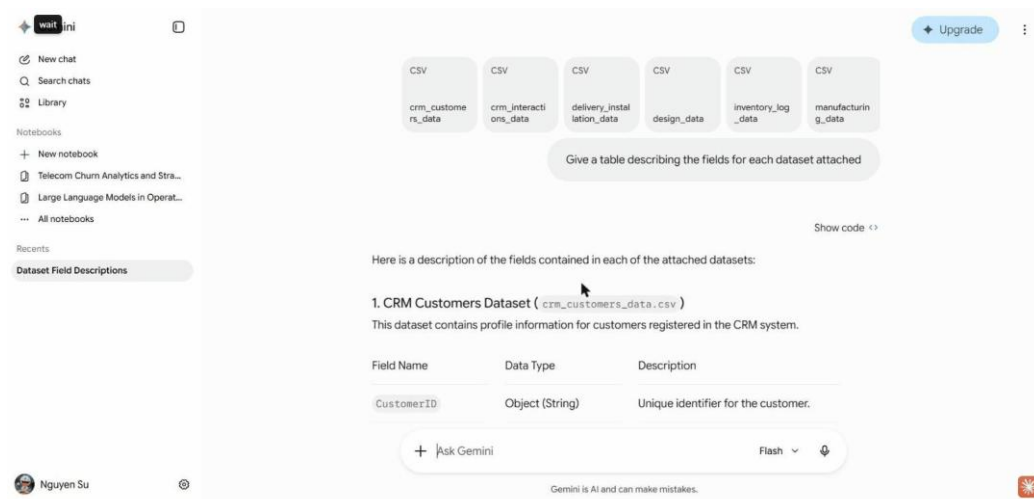


Figure 1 - Gemini describes the fields of each attached dataset - a quick way to learn an unfamiliar schema.

Step 2 - Ask Gemini to build the app (Canvas) (Slide 13)

Now turn the data into a tool. Canvas is Gemini's build mode - it writes and runs a small web app beside the chat.

1. Click the + button and turn on Canvas (the input shows "Let's write or build together").
2. Enter the build prompt below and send it.

Build a simple app with a dashboard. The app should allow me to upload the CRM data (the csv file above) and provide a number of visualisations to help me understand the sales performance.

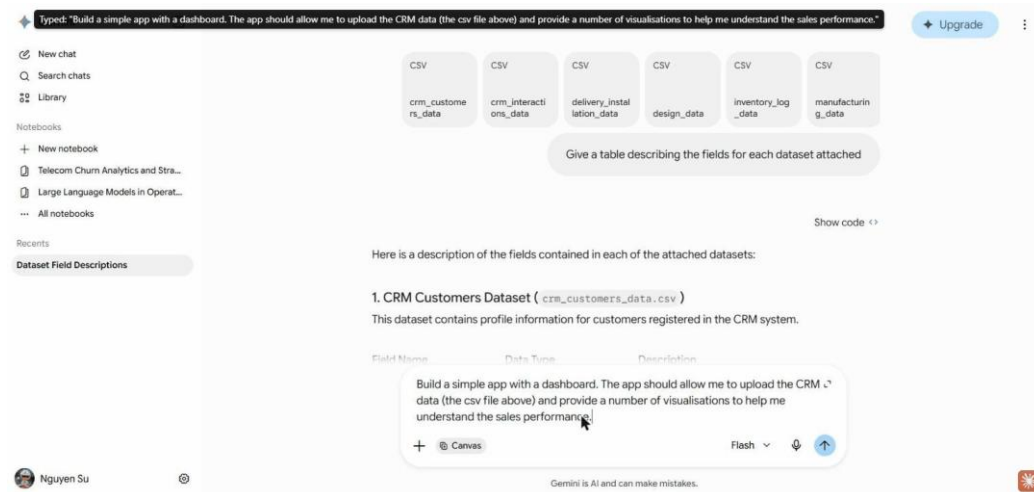


Figure 2 - With Canvas on, a single plain-English prompt asks Gemini to build a sales-performance dashboard app.

Tip: Canvas takes a minute or two to write the app - it generates a full React application, not just a chart.

Step 3 - View the dashboard (Slides 14-15)

When it finishes, the Preview pane shows a working dashboard. Gemini built an "ApexCRM Analytics" app with KPI cards (Closed Revenue, Pipeline Value, Win Rate, Avg Deal Size, Cycle Velocity), interactive filters, and an Upload Files button so you can load your own CSV.

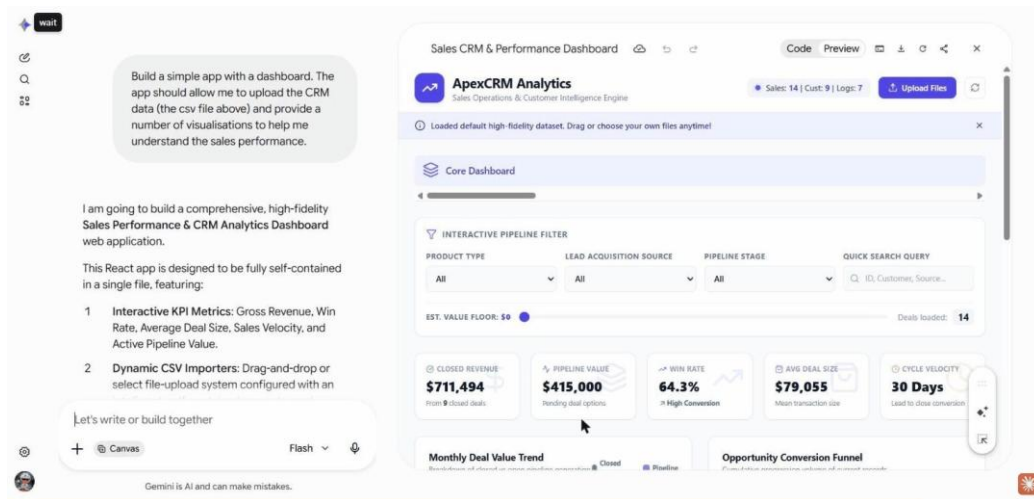


Figure 3 - The generated dashboard: KPI metrics across the top with an interactive pipeline filter and a CSV uploader.

Scroll the app to see the visualisations - a Monthly Deal Value Trend, an Opportunity Conversion Funnel, Product Distribution, and Lead Source Velocity - all generated from the data.

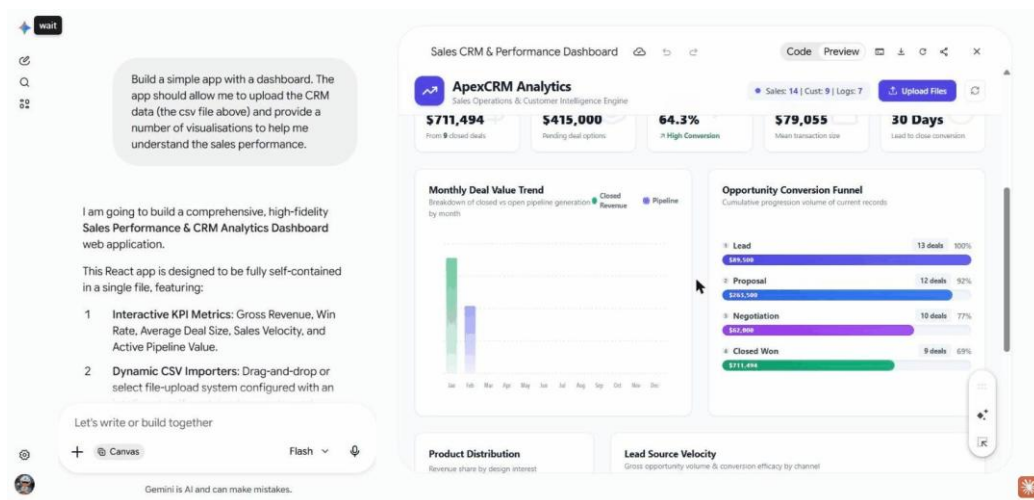


Figure 4 - The charts Gemini created to explain sales performance, with no manual coding.

Tip: the app loads a sample dataset so you see it working immediately. Click Upload Files to load your own CRM CSV (e.g. sales_data.csv). You can also open the Code tab to read the generated React/HTML if you're curious.

Step 4 - Fine-tune a chart by asking (Slide 16)

The app is not fixed - you refine it by asking, the same way you built it. Send a follow-up message in the Canvas chat describing the change you want. Here we change one chart's type.

Change the Product Distribution donut chart into a bar chart instead.

Gemini edits the app's code and re-renders only that panel. The donut becomes a horizontal bar chart - same data, clearer ranking of each product line.

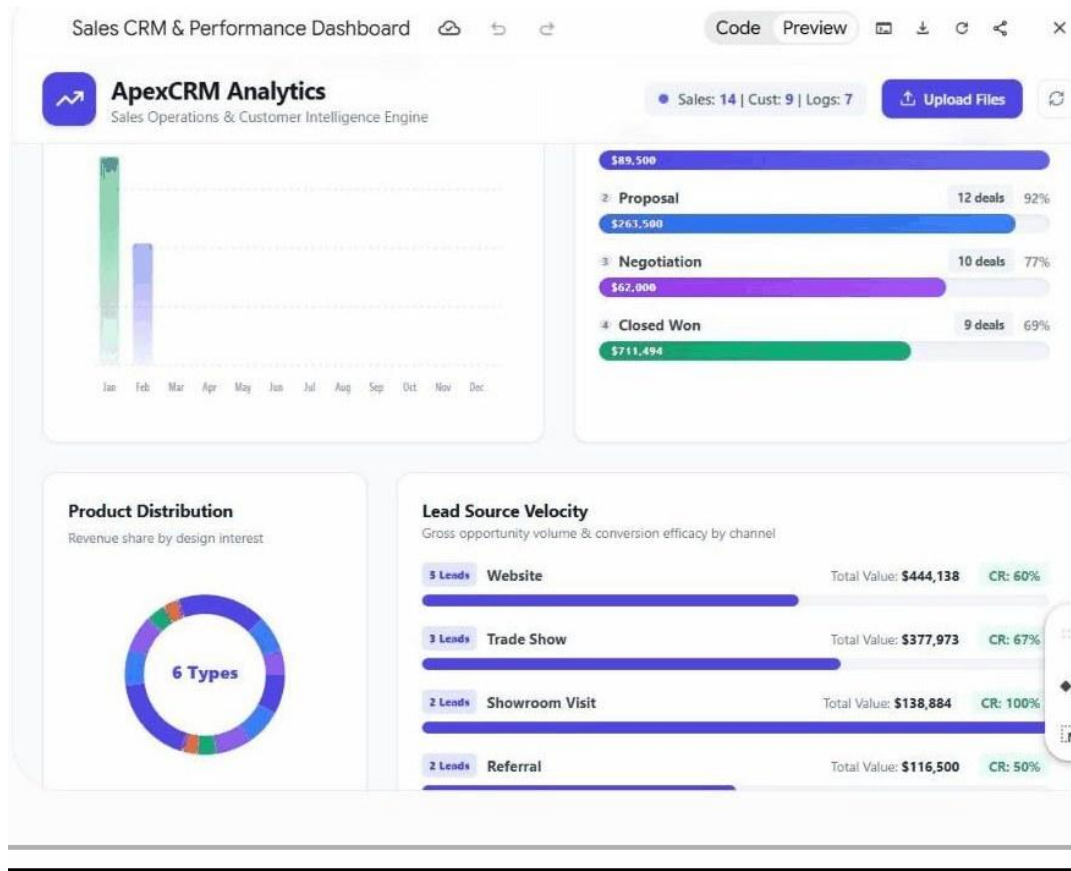


Figure 5 - Before: Product Distribution drawn as a donut chart (6 Types).

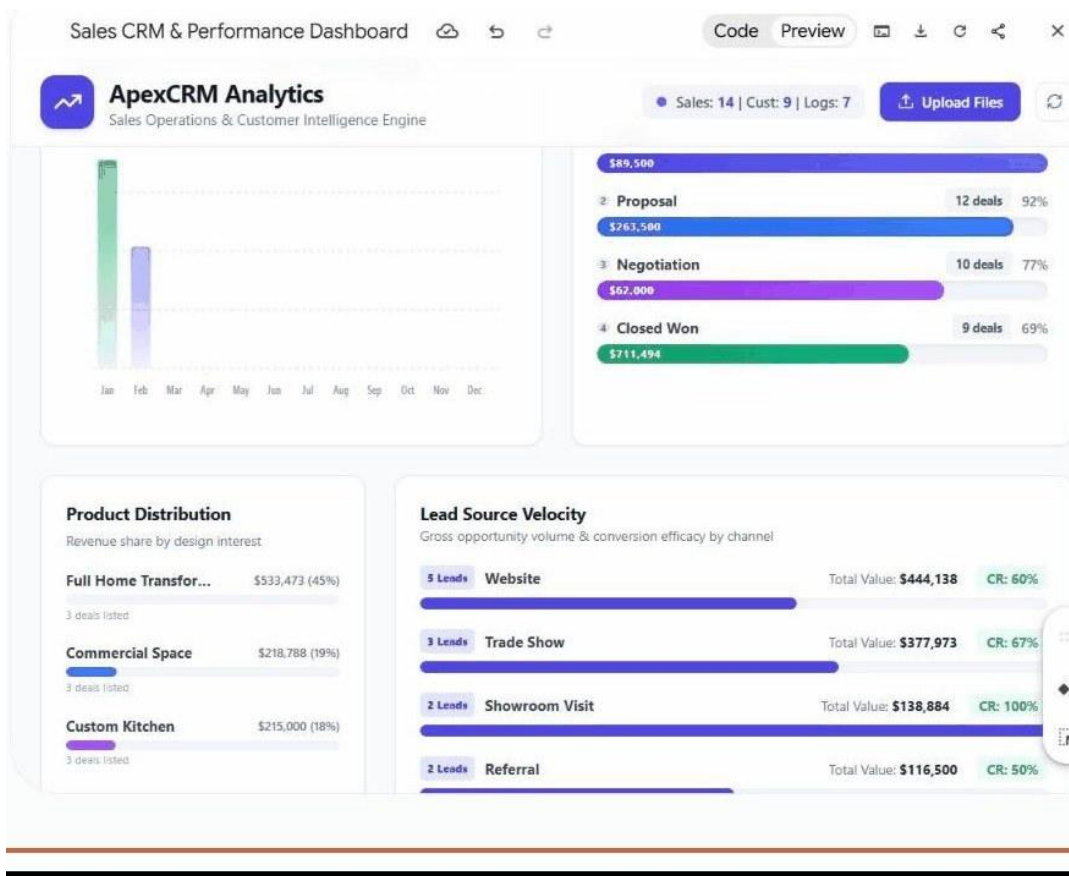


Figure 6 - After asking Gemini: the same panel is now a ranked horizontal bar chart - Full Home Transformation \$533,473 (45%), Commercial Space (19%), Custom Kitchen (18%).

Tip: you can also use select-and-ask - click an element in the Preview, then type the change you want to that specific element.

Step 5 - Add an AI feature: a natural-language filter (Slide 17)

You can add genuinely new capability in plain English. Ask Gemini to build a search bar that interprets a question and filters the dashboard.

Add a search bar at the top of the dashboard where the user can type a question about the data in plain English, and the app interprets it and applies filters to the dashboard. For example: revenue from April to September for Full Home Transformation. Show the active filter and update the metrics accordingly.

Gemini rebuilds the app with a "Natural Language Intelligence" bar (an Ask AI button and example chips). Test it: type a question and click Ask AI.

1. Type a question, e.g. "revenue from April to September for Full Home Transformation".
2. Click Ask AI (or pick a suggested chip).

3. The app interprets the question and applies the matching filters to the whole dashboard.

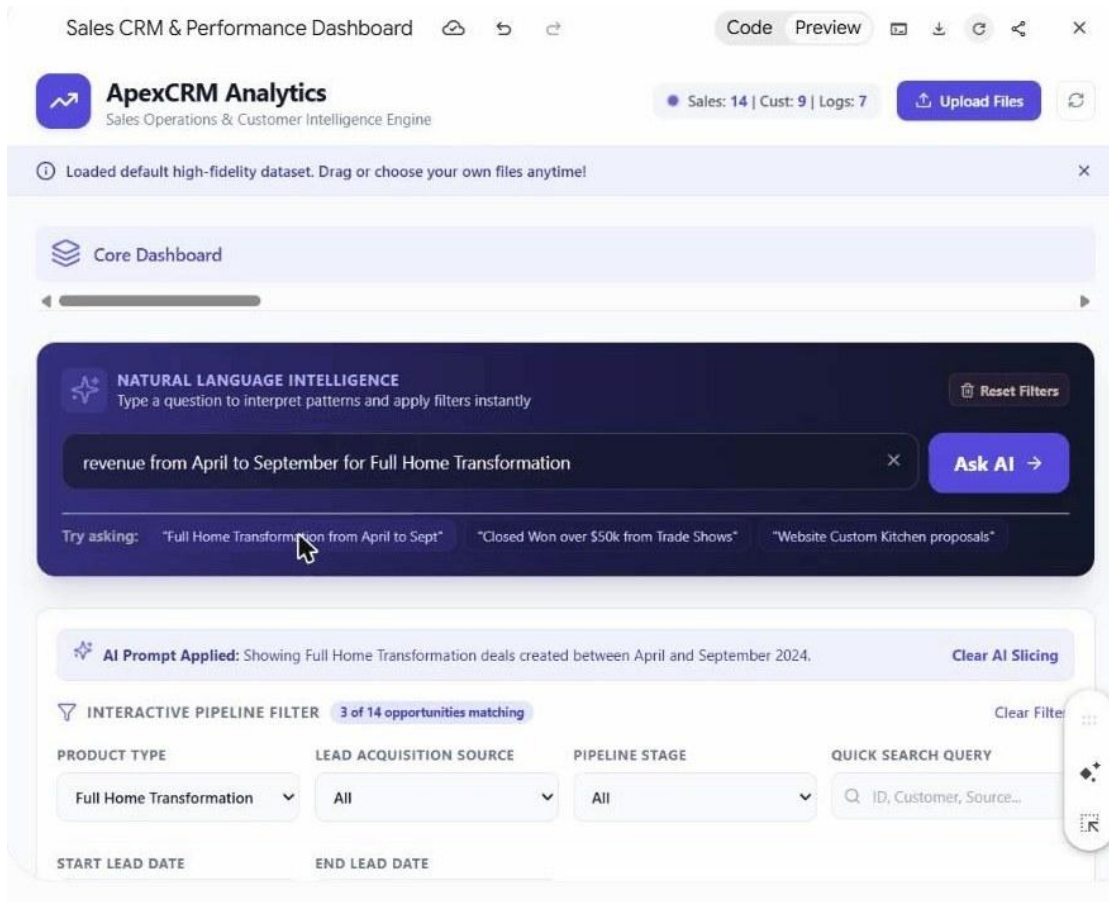


Figure 7 - The natural-language filter in action: the question is parsed into Showing Full Home Transformation deals created between April and September 2024, the Product Type filter is set automatically, and the pipeline narrows to 3 of 14 matching opportunities.

Tip: use Clear AI Slicing or Reset Filters to return to the full dataset. The bar parses the question with Gemini, so phrasing it like a plain question works best.

Good to know

- Upload the data first and ask for a field table - it is the fastest way to understand an unfamiliar dataset before building anything.
- Canvas builds a real app (a React web app), not just a chart; give it a clear goal and let it run.
- You refine the app the same way you built it - in plain English. There is no separate 'edit mode' to learn.
- Select-and-ask: click a specific element in the preview, then describe the change you want to that element.
- Generated apps can vary run to run; if a build looks off, ask Gemini to adjust it or rebuild. You can open the Code tab to inspect or export what it wrote.